For Examiner's Use

2 At t = 0.0 s, a golf ball is launched with initial velocity u and angle θ to the horizontal as shown in Fig. 2.1. After 8.8 s, it lands at a distance of 220 m away at the same level where it was launched.

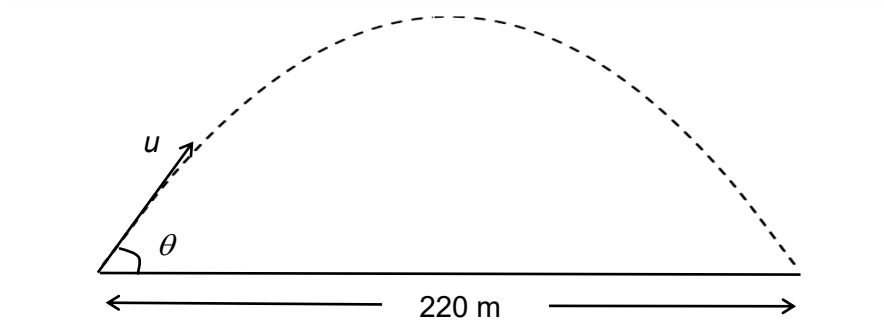

Fig. 2.1

(a) Determine u and θ.

u = ... m s^{-1}

θ = ... ° [5]

(b) **(i)** Calculate magnitude of the *change in velocity*, Δv, of the golf ball in the duration t = 0.0 s to t = 7.0 s.

Δv = ... m s^{-1} [1]

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(ii) By scale drawing, draw the vectors representing u and ΔV on Fig. 2.2. Hence, determine the magnitude and direction of the velocity v of the ball at 7.0 s. Label the vectors clearly.

Fig. 2.2

magnitude of v = …… m s^{-1}

angle below horizontal = …… ° [3]

[Turn over